

IoT, Cloud and Machine Learning: The Building Blocks

This article touches upon the building blocks that are necessary to enable machine learning in the data received from IoT, and how cloud infrastructure can help if we use the power of open source tools effectively.

MQTT (<http://mqtt.org/>) or Message Queuing Telemetry Transport is an open OASIS and ISO standard (ISO/IEC 20922), lightweight, publish-subscribe network protocol that transports messages between devices. The protocol usually runs over TCP/IP; however, any network protocol that provides ordered, lossless, bi-directional connections can support MQTT. It is designed for connections with remote locations where a 'small code footprint' is required or the network bandwidth is limited (Source: <https://en.wikipedia.org/wiki/MQTT>).

Figure 1 explains how to connect the device to the cloud. Let's discuss the components in detail in the sections below.

MQTT broker

MQTT broker acts as a common point for receiving and publishing messages from clients who are subscribed with it. Clients can connect to the broker, and then receive messages from and also publish messages to the topics. In our case, clients are at the IoT device side and broker resides at the cloud virtual machine. So, the cloud MQTT broker receives data from IoT devices through topics, where devices

publish the messages. Also, the cloud can communicate with devices by publishing messages to topics that are subscribed by the device.

We have a number of options here, and one popular choice is Mosquitto; you can download it from <https://mosquitto.org/download/>.

Eclipse Mosquitto is an open source (EPL/EDL licensed) message broker that implements the MQTT protocol versions 5.0, 3.1.1 and 3.1. It is lightweight, and is suitable for use on devices ranging from low power, single-board computers to full servers. The Mosquitto project also provides a C library for implementing MQTT clients, as well as the very popular `mosquitto_pub` and `mosquitto_sub` command line MQTT clients (Source: <https://mosquitto.org/>).

Eclipse Paho is another option. You can get more details at <https://www.eclipse.org/paho/>.

MQTT client

MQTT client acts as a client connected to the broker. It can receive and publish messages from/to topics that are subscribed by the client from the IoT devices. We have a number of options for MQTT client, and we have already mentioned the Mosquitto C library as MQTT client in the earlier section (<https://github.com/eclipse/mosquitto>). Eclipse Paho client for C is another option (<https://www.eclipse.org/paho/clients/c/>).

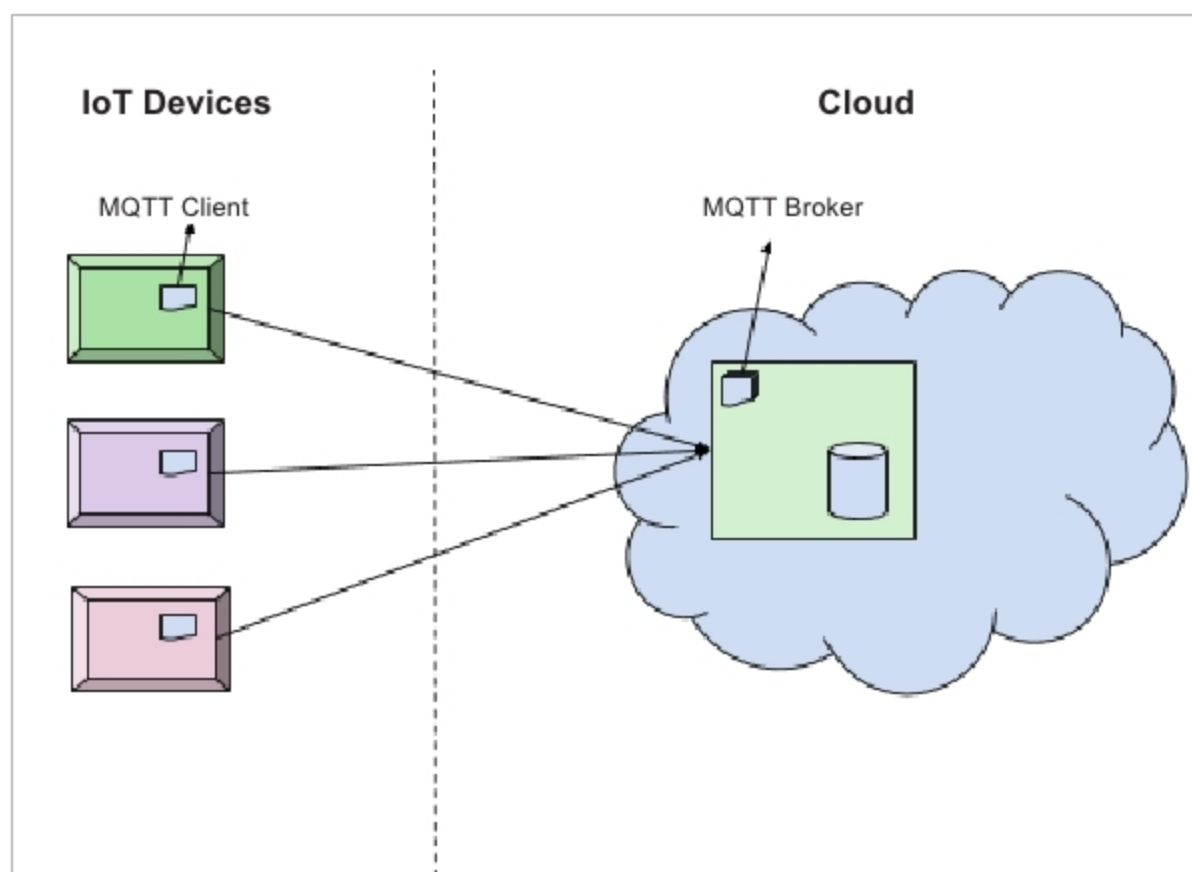


Figure 1: Connecting the device to the cloud

